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scikit-eo: A Python package for Remote Sensing Data Analysis

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Summary

In recent years, a growing body of space-borne and drone imagery has become available with increasing spatial and temporal resolutions. This remotely sensed data has enabled researchers to address and tackle a broader range of challenges effectively by using novel tools and data. However, analysts spend an important amount of time finding the adequate libraries to read and process remotely sensed data.

With an increasing amount of open access data, there is a growing need to account for effective open source tools to read, process and execute analysis that contributes to underpin patterns, changes and trends that are critical for environmental studies. Applications that integrate spatial-temporal data are used to study a variety of complex environmental processes, such as monitoring and assessment of land cover changes (Chaves et al., 2020), crop classifications (Pott et al., 2021), deforestation (Tarazona et al., 2018), impact on urbanization level (Trinder & Liu, 2020), climate change impacts (Yang et al., 2013) including assessments of glacier retreat (Hugonnet et al., 2018) and related hydrological change (Huss & Hock, 2018), biodiversity conservation (Cavender-Bares et al., 2022), and disaster management (Kucharczyk & Hugenholtz, 2021).

To bridge the gaps in remotely sensed data processing tools, we here introduce **scikit-eo**, a brand-new Python package for satellite remote sensing analysis. Unlike other tools, it is a centralized, scalable and open-source toolkit. **scikit-eo** stands out with its unique ability to be used in various settings, from a lecturer room to a crucial part of any Python environment in a research project. The majority of the tools included in **scikit-eo** are derived from peer-reviewed scientific publications, ensuring their reliability and accuracy.

By integrating this diverse set of tools, **scikit-eo** allows to focus on analyzing the results of data rather than being bogged down by complex lines of code. With its centralized structure, integrated use cases, and example data, **scikit-eo** empowers to optimize resources and dedicate more attention to the meaningful interpretation of findings in a more efficient way.

Statement of Need

As remote sensing data and sophisticated processing tools become increasingly available, there is a growing need for scalable and customized toolkits to help environmental researchers classify satellite and drone imagery quickly and accurately. Our goal is to simplify the identification of patterns, changes, and trends that are crucial for environmental research and analysis.

In this paper, we introduce **scikit-eo**, a specialized library with tailored analysis capabilities designed to meet the unique demands of environmental studies. From statistical methods to machine learning algorithms, **scikit-eo** assists researchers in uncovering intricate spatial patterns, relationships, and trends, while also simplifying the evaluation and calibration of generated outputs.

scikit-eo is an open-source package built entirely in Python through Object-Oriented Programming and Structured Programming that provides a helpful variety of remote sensing tools (see Figure 1), from primary and exploratory functions to more advanced methods to classify, calibrate, or fuse satellite imagery. Depending on the users' needs, **scikit-eo** can provide the basic but essential land cover characterization mapping, including the confusion matrix and the required metrics such as user's accuracy, producer's accuracy, omission and commission errors. These required metrics can be combined as a pandas `DataFrame` object. Furthermore, a class prediction map is a result of land cover mapping, i.e., a land cover map, which represents the output of the classification algorithm or the output of the segmentation algorithm. These two outcomes must include uncertainties with a confidence level (e.g., at 95% or 90%). All required metrics from the confusion matrix can be easily computed and included confidence levels with **scikit-eo** following guidance proposed by Olofsson et al. (2014). Other useful tools for remote sensing analysis can be found in this package; and for more information about the full list of the supported functions, tutorials as well as how to install and execute the package within a Python setting, visit the scikit-eo website.

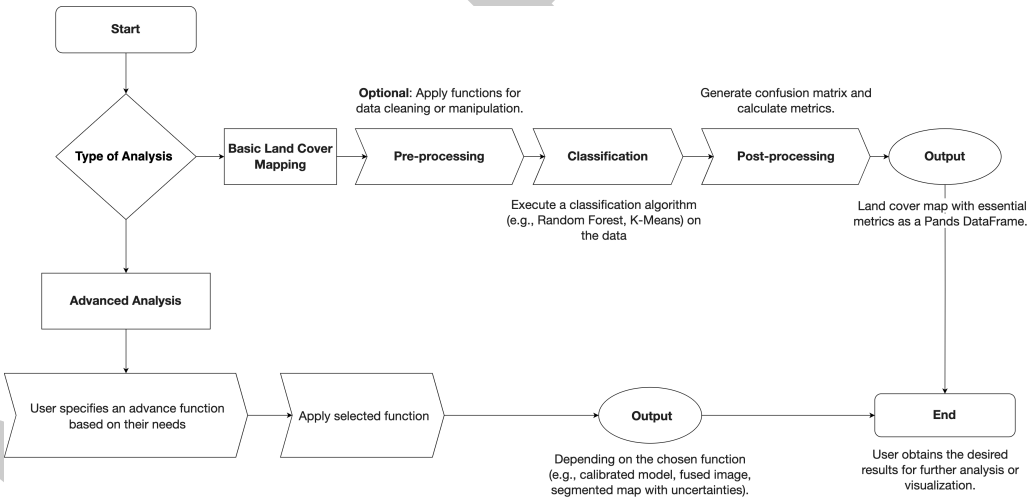

Figure 1: Workflow of main functionalities of the *scikit-eo* python package as well as outputs that can be obtained by using the tools developed.

Audience

scikit-eo is an adaptable Python package that covers multiple users, including students, remote sensing professionals, environmental analysis researchers, and organizations looking for satellite image analysis. Its comprehensive features make it well-suited for various applications, such as university teaching, that includes technical and practical sessions and cutting-edge research using the most recent machine learning and deep learning techniques applied to remote sensing whether the users are students seeking insights from a satellite image analysis or an experienced researcher looking for advanced tools, **scikit-eo** offers a valuable resource to support the most valuable methods for environmental studies.

scikit-eo as a research tool:

In environmental studies, particularly in land cover classification, the availability of scalable but user-friendly software tools is crucial for facilitating research, analysis, and modelling. With the widespread adoption of Python as one of the most popular programming languages, particularly in the GIScience and remote sensing fields, developing specialized packages has massively enhanced the effectiveness of environmental research. **scikit-eo** is a dedicated piece of software tailored to address unique challenges encountered in land cover mapping, forest or land degradation (Tarazona & Miyasiro-López, 2020), and the fusion of multiple satellite imagery from several formats (Tarazona et al., 2021). **scikit-eo** provides the assessment and calibration metrics to evaluate the provided outputs. The current version **scikit-eo** requires that users provide the dataset to process. However, we expect to provide a wide range of functionalities for acquiring environmental data from diverse sources and file formats, enabling researchers to access satellite imagery.

One of **scikit-eo's** key strengths is its advanced analysis capabilities. It provides a rich suite of algorithms specifically designed for environmental studies, including statistical analysis, machine learning, deep learning, data fusion, and spatial analysis. Researchers can leverage these tools to explore patterns, relationships, and trends within their datasets, uncover complex land or forest degradation or mapping, classify the land cover, and generate insightful visualizations.

scikit-eo in the lecture room:

scikit-eo can be part of a lecturer room as part of the set of tools for environmental studies where a quantitative approach and computer labs are required. Therefore, first of all an appropriate introduction of Python, the basics of remote sensing, and the relevance of environmental studies to address climate change challenges or impacts of anthropogenic activity are needed. Lectures can use the simplicity of **scikit-eo** routines to execute supervised classification methods, `Principal Components Analysis`, `Spectral Mixture Analysis`, `Mapping forest` or `land degradation` and more types of analysis. By reducing the number of required lines of code, students can focus on the analysis and how the methods work rather than dealing with complex and unnecessary programming tasks. Lecturers can structure their computer labs using open data sources and integrate **scikit-eo** to allow students to understand the importance of the calibration and assessment metrics, get insights about the classification mapping using satellite imagery, and provide an introduction to more advanced methods that include machine learning techniques.

scikit-eo as open source tool:

As open-source software keeps transforming the landscape of scientific research (Community et al., 2019), enabling collaboration, reproducibility and transparency, **scikit-eo** was explicitly developed as an open-source tool. **scikit-eo** integrates most of the popular open source python libraries from the so-called geo-python stack for remote sensing (e.g.,. numpy, pandas, rasterio and few more) to extent and create a centralised package for advanced spatial analysis of remotely sensed data. The package provides researchers and developers with a free, scalable, and community-driven platform to process, analyse, and visualise satellite imagery, specifically, centralising multiple functions' use for classification and mapping land cover.

Functionalities

Main tools

Scikit-eo includes several algorithms to process satellite images to assess complex environmental processes and impacts. These include functions such as atmospheric correction, machine learning and deep learning techniques, estimating area and uncertainty, linear trend analysis,

combination of optical and radar images, and classification sub-pixel, to name a few. Some main functions are listed below:

Table 1: Main tools available for **scikit-eo** package.

Name of functions/classes	Description
`mla`	Supervised Classification in Remote Sensing
`calmla`	Calibrating Supervised Classification in Remote Sensing
`confintervalML`	Information of Confusion Matrix by proportions of area, overall accuracy, user's accuracy with confidence interval and estimated area with confidence interval as well.
`deepLearning`	Deep Learning algorithms
`atmosCorr`	Radiometric and Atmospheric Correction
`rkmeans`	K-means classification
`calkmeans`	This function allows to calibrate the kmeans algorithm. It is possible to obtain the best k value and the best embedded algorithm in kmeans.
`pca`	Principal Components Analysis
`linearTrend`	Linear trend is useful for mapping forest degradation or land degradation
`fusionrs`	This algorithm allows to fuse images coming from different spectral sensors (e.g., optical-optical, optical and SAR or SAR-SAR). Among many of the qualities of this function, it is possible to obtain the contribution (%) of each variable in the fused image
`sma`	Spectral Mixture Analysis - Classification sup-pixel
`tassCap`	The Tasseled-Cap Transformation

For more information the reader is referred to the scikit-eo website.

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